

JEE Mains 2019 Chapter wise Question Bank

Differentiation – Questions

Q1

The value of $\int_0^{\pi} |\cos x|^3 dx$ is:

- (1) 0 (2) $\frac{4}{3}$
 (3) $\frac{2}{3}$ (4) $-\frac{4}{3}$

9 Jan Morning

Q2

If $x = 3 \tan t$ and $y = 3 \sec t$, then the value of $\frac{d^2y}{dx^2}$ at $t = \frac{\pi}{4}$, is:

- (1) $\frac{1}{3\sqrt{2}}$ (2) $\frac{1}{6\sqrt{2}}$
 (3) $\frac{3}{2\sqrt{2}}$ (4) $\frac{1}{6}$

9 Jan Evening

Q3

Let $f: \mathbf{R} \rightarrow \mathbf{R}$ be a function such that

$f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$, $x \in \mathbf{R}$. Then $f(2)$ equals:

- (1) -4 (2) 30
 (3) -2 (4) 8

10 Jan Morning

Q4

If $x \log_e (\log_e x) - x^2 + y^2 = 4$ ($y > 0$), then $\frac{dy}{dx}$ at $x = e$ is equal to :

- (1) $\frac{(1+2e)}{2\sqrt{4+e^2}}$ (2) $\frac{(2e-1)}{2\sqrt{4+e^2}}$
 (3) $\frac{(1+2e)}{\sqrt{4+e^2}}$ (4) $\frac{e}{\sqrt{4+e^2}}$

11 Jan Morning

Q5

For $x > 1$, if $(2x)^{2y} = 4e^{2x-2y}$, then $(1 + \log_e 2x)^2 \frac{dy}{dx}$ is equal to :

- (1) $\frac{x \log_e 2x - \log_e 2}{x}$ (2) $\log_e 2x$
 (3) $\frac{x \log_e 2x + \log_e 2}{x}$ (4) $x \log_e 2x$

12 Jan Morning

Q6

Let f be a differentiable function such that $f(1) = 2$ and $f'(x) = f(x)$ for all $x \in \mathbf{R}$. If $h(x) = f(f(x))$, then $h'(1)$ is equal to :

- (1) $2e^2$ (2) $4e$
 (3) $2e$ (4) $4e^2$

12 Jan Evening

Q7

If $2y = \left(\cot^{-1} \left(\frac{\sqrt{3} \cos x + \sin x}{\cos x - \sqrt{3} \sin x} \right) \right)^2$, $x \in \left(0, \frac{\pi}{2} \right)$ then $\frac{dy}{dx}$

is equal to :

- (1) $\frac{\pi}{6} - x$ (2) $x - \frac{\pi}{6}$
 (3) $\frac{\pi}{3} - x$ (4) $2x - \frac{\pi}{3}$

8 April Morning

Q8

If $f(1) = 1, f'(1) = 3$, then the derivative of $f(f(f(x))) + (f(x))^2$ at $x = 1$ is :

- (1) 33 (2) 12 (3) 15 (4) 9

8 April Evening

Q9

If $e^y + xy = e$, the ordered pair $\left(\frac{dy}{dx}, \frac{d^2y}{dx^2} \right)$ at $x = 0$ is

equal to :

- (1) $\left(\frac{1}{e}, -\frac{1}{e^2} \right)$ (2) $\left(-\frac{1}{e}, \frac{1}{e^2} \right)$
 (3) $\left(\frac{1}{e}, \frac{1}{e^2} \right)$ (4) $\left(-\frac{1}{e}, -\frac{1}{e^2} \right)$

12 April Morning

Q10

The derivative of $\tan^{-1} \left(\frac{\sin x - \cos x}{\sin x + \cos x} \right)$, with respect to $\frac{x}{2}$,

where $\left(x \in \left(0, \frac{\pi}{2} \right) \right)$ is :

- (1) 1 (2) $\frac{2}{3}$ (3) $\frac{1}{2}$ (4) 2

12 April Evening

JEE Mains 2019 Chapter wise Question Bank

Differentiation - Solutions

Q1

$$\begin{aligned}
 (2) \quad I &= \int_0^{\pi} |\cos x|^3 dx \\
 &= 2 \int_0^{\pi/2} \cos^3 x dx \\
 &= \frac{2}{4} \int_0^{\pi/2} (3 \cos x + \cos 3x) dx \\
 &\quad [\because \cos 3\theta = 4\cos^3 \theta - 3\cos \theta] \\
 &= \frac{1}{2} \left[3 \sin x + \frac{\sin 3x}{3} \right]_0^{\pi/2} \\
 &= \frac{1}{2} \left(3 - \frac{1}{3} \right) = \frac{4}{3}
 \end{aligned}$$

9 Jan Morning

Q2

$$(2) \because x = 3 \tan t \Rightarrow \frac{dx}{dt} = 3 \sec^2 t$$

$$\text{and } y = 3 \sec t \Rightarrow \frac{dy}{dt} = 3 \sec t \cdot \tan t$$

$$\therefore \frac{dy}{dx} = \frac{dy/dt}{dx/dt} \quad \therefore \frac{dy}{dx} = \frac{\tan t}{\sec t} = \sin t$$

$$\therefore \frac{d^2 y}{dx^2} = \frac{d}{dt} (\sin t) \cdot \frac{dt}{dx}$$

$$= \cos t \cdot \frac{1}{3 \sec^2 t}$$

$$\begin{aligned}
 \therefore \frac{d^2 y}{dx^2} \left(\text{at } t = \frac{\pi}{4} \right) &= \frac{1}{3} \cdot \left(\frac{1}{\sqrt{2}} \right)^3 \\
 &= \frac{1}{6\sqrt{2}}
 \end{aligned}$$

9 Jan Evening

Q3

$$(3) \text{ Let } f(x) = x^3 + ax^2 + bx + c$$

$$f'(x) = 3x^2 + 2ax + b \Rightarrow f'(1) = 3 + 2a + b$$

$$f''(x) = 6x + 2a \Rightarrow f''(2) = 12 + 2a$$

$$f'''(x) = 6 \Rightarrow f'''(3) = 6$$

$$\therefore f(x) = x^3 + f'(1)x^2 + f''(2)x + f'''(3)$$

$$\therefore f'(1) = a \Rightarrow 3 + 2a + b = a \Rightarrow a + b = -3$$

...(1)

$$\text{also } f''(2) = b \Rightarrow 12 + 2a = b \Rightarrow 2a - b = -12$$

...(2)

$$\text{and } f'''(3) = c \Rightarrow c = 6$$

Add (1) and (2)

$$3a = -15 \Rightarrow a = -5 \Rightarrow b = 2$$

$$\Rightarrow f(x) = x^3 - 5x^2 + 2x + 6$$

$$\Rightarrow f(2) = 8 - 20 + 4 + 6 = -2$$

10 Jan Morning

Q4

Differentiation

(2) Consider the equation,

$$x \log_e (\log_e x) - x^2 + y^2 = 4$$

Differentiate both sides w.r.t. x ,

$$\log_e (\log_e x) + x \cdot \frac{1}{x \cdot \log_e x} - 2x + 2y \frac{dy}{dx} = 0$$

$$\log_e (\log_e x) + \frac{1}{\log_e x} - 2x + 2y \frac{dy}{dx} = 0 \quad \dots(1)$$

When $x = e$, $y = \sqrt{4 + e^2}$. Put these values in (1),

$$0 + 1 - 2e + 2\sqrt{4 + e^2} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{2e - 1}{2\sqrt{4 + e^2}}.$$

11 Jan Morning

Q5

(1) Consider the equation,

$$(2x)^{2y} = 4e^{2x-2y}$$

Taking log on both sides

$$2y \ln(2x) = \ln 4 + (2x - 2y) \quad \dots(1)$$

Differentiating both sides w.r.t. x ,

$$2y \frac{1}{2x} + 2 \ln(2x) \frac{dy}{dx} = 0 + 2 - 2 \frac{dy}{dx}$$

$$2 \frac{dy}{dx} (1 + \ln(2x)) = 2 - \frac{2y}{x} = \frac{2x - 2y}{x} \quad \dots(2)$$

From (1) and (2),

$$\frac{dy}{dx} (1 + \ln 2x) = 1 - \frac{1}{x} \left(\frac{\ln 2 + x}{1 + \ln 2x} \right)$$

$$\Rightarrow (1 + \ln 2x)^2 \frac{dy}{dx} = 1 + \ln(2x) - \left(\frac{x + \ln 2}{x} \right)$$

$$= \frac{x \ln(2x) - \ln 2}{x}$$

12 Jan Morning

JEE Mains 2019 Chapter wise Question Bank

Q6

(2) Since, $f'(x) = f(x)$

$$\text{Then, } \frac{f'(x)}{f(x)} = 1$$

$$\Rightarrow \frac{f'(x)}{f(x)} = dx \Rightarrow \frac{f'(x)}{f(x)} dx = \int dx$$

$$\Rightarrow \ln |f(x)| = x + c$$

$$f(x) = \pm e^{x+c} \quad \dots(1)$$

Since, the given condition

$$f(1) = 2$$

From eqⁿ (1) $f(x) = e^{x+c} = e^c e^x$

$$\text{Then, } f(1) = e^c \cdot e^1$$

$$\Rightarrow 2 = e^c \cdot e$$

$$\Rightarrow \frac{2}{e} = e^c$$

Then, from eqⁿ (1)

$$f(x) = \frac{2}{e} e^x$$

$$\Rightarrow f'(x) = \frac{2}{e} e^x$$

$$\text{Now } h(x) = f(f(x))$$

$$\Rightarrow h'(x) = f'(f(x)) \cdot f'(x)$$

$$h'(1) = f'(2) \cdot f'(1) = \frac{2}{e} e^2 \cdot \frac{2}{e} e = 4e$$

12 Jan Evening

Q7

Differentiation

(Bouns)

(none)

$$2y = \left[\cot^{-1} \left(\frac{\frac{\sqrt{3}}{2} \cos x + \frac{1}{2} \sin x}{\frac{1}{2} \cos x - \frac{\sqrt{3}}{2} \sin x} \right) \right]^2$$

$$\Rightarrow 2y = \left[\cot \left(\frac{\cos\left(\frac{\pi}{6}\right)}{\sin\left(-\right)} \right) \right]^2$$

$$\Rightarrow 2y = \left[\cot^{-1} \left(\cot \left(\frac{\pi}{6} - x \right) \right) \right]^2 \because \frac{\pi}{6} - x \in \left(-\frac{\pi}{3}, \frac{\pi}{6} \right)$$

$$\Rightarrow 2y = \begin{cases} \left(\frac{7\pi}{6} - x \right)^2, & \text{if } \frac{\pi}{6} - x \in \left(-\frac{\pi}{3}, 0 \right) \\ \left(\frac{\pi}{6} - x \right)^2, & \text{if } \frac{\pi}{6} - x \in \left(0, \frac{\pi}{6} \right) \end{cases}$$

$$\Rightarrow \frac{dy}{dx} = \begin{cases} x - \frac{7\pi}{6} & \text{if } x \in \left(\frac{\pi}{6}, \frac{\pi}{2} \right) \\ x - \frac{\pi}{6} & \text{if } x \in \left(0, \frac{\pi}{6} \right) \end{cases}$$

8 April Morning

Q8

(1) Let $g(x) = f(f(f(x))) + (f(x))^2$

Differentiating both sides w.r.t. x , we get

$$\begin{aligned} g'(x) &= f'(f(f(x)))f'(f(x))f'(x) + 2f(x)f'(x) \\ g'(1) &= f'(f(f(1)))f'(f(1))f'(1) + 2f(1)f'(1) \\ &= f'(f(1))f(1)f'(1) + 2f(1)f'(1) \\ &= 3 \times 3 \times 3 + 2 \times 1 \times 3 = 27 + 6 = 33 \end{aligned}$$

8 April Evening

Q9

JEE Mains 2019 Chapter wise Question Bank

(2) Given, $e^y + xy = e$... (i)

Putting $x = 0$ in (i), $\Rightarrow e^y = e \Rightarrow y = 1$

On differentiating (i) w. r. to x

$$e^y \frac{dy}{dx} + x \frac{dy}{dx} + y = 0 \quad \dots (ii)$$

Putting $y = 1$ and $x = 0$ in (ii),

$$e \frac{dy}{dx} + 0 + 1 = 0 \Rightarrow \frac{dy}{dx} = -\frac{1}{e}$$

On differentiating (ii) w. r. to x ,

$$e^y \frac{d^2y}{dx^2} + \frac{dy}{dx} \cdot e^y \cdot \frac{dy}{dx} + x \frac{d^2y}{dx^2} + \frac{dy}{dx} + \frac{dy}{dx} = 0 \quad \dots (iii)$$

Putting $y = 1, x = 0$ and $\frac{dy}{dx} = -\frac{1}{e}$ in (iii),

$$e \frac{d^2y}{dx^2} + \frac{1}{e} - \frac{2}{e} = 0 \Rightarrow \frac{d^2y}{dx^2} = \frac{1}{e^2}$$

$$\text{Hence, } \left(\frac{dy}{dx}, \frac{d^2y}{dx^2} \right) \equiv \left(-\frac{1}{e}, \frac{1}{e^2} \right)$$

12 April Morning

Q10

$$\begin{aligned} (4) f(x) &= \tan^{-1} \left(\frac{\tan x - 1}{\tan x + 1} \right) \\ &= -\tan^{-1} \left(\tan \left(\frac{\pi}{4} - x \right) \right) \quad \left[\because \frac{\pi}{4} - x \in \left(-\frac{\pi}{4}, \frac{\pi}{4} \right) \right] \end{aligned}$$

$$\text{So, } f(x) = -\left(\frac{\pi}{4} - x \right) = x - \frac{\pi}{4}$$

$$\text{Let } y = \frac{x}{2} \Rightarrow f(y) = 2y - \frac{\pi}{4}$$

$$\text{Now, differentiate w.r.t. } y, \quad \frac{df(y)}{dy} = 2.$$

12 April Evening